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Team Details

Team ID : [Enter Team ID]

Team Name : [Enter Team Name]

Leader Name : [Enter Leader Name]

Domain Chosen : **Web & Product Development**

Problem Statement :

Health Outbreak Detection & Alert System

Our Solution: **VARSHA** — *Vector-And-Rain-driven Surveillance for Health Alerts*

Problem Analysis *Mumbai's monsoon outbreaks are predictable. The response isn't.*

Real-world issue

Every monsoon, floods spread leptospirosis, dengue and malaria across Mumbai.

- **66 suspected deaths** from these 3 diseases in 2025 (BMC review)
- **8x jump** in leptospirosis after the 26 July 2005 deluge
- **July 2026:** leptospirosis cases more than doubled, **33 → 78**, after a week of heavy rain

Why It Keeps Happening

People catch leptospirosis on predictable days: when they waded through floodwater. BMC says preventive care must start **within 72 hours**. But today:

- Advice is **citywide** and comes **days after** the flood
- Official outbreak reports go online **~6 weeks later**
- AI tools read the news only **after people fall sick**

Result: the warning comes after the window has closed.

WHO IS AFFECTED

- Families in flood-prone slums**
About 4 in 10 Mumbaikars live in slums, many next to chronic flood spots.
- BMC health teams**
Must spread limited fever camps, doctors and spray teams over 24 wards, without knowing where cases will hit.
- Hospitals & blood banks**
Get sudden fever surges; dengue patients need platelets that last only 5 days.

DAY 0 HEAVY RAIN

CLOUDBURST
Up to ~944 mm of rain in a single day (26 Jul 2005).

FLOODED STREETS
BMC lists 386 flood spots (2022); people waded through to get home.

DAY 0-3 EXPOSURE

INFECTION
Germs from rat urine in floodwater enter through cuts and wet skin.

72-HOUR WINDOW
BMC advice: start preventive care within 24-72 hours.

DAY 3-12 SILENT SPREAD

- NOBODY IS WARNED**
- No symptoms yet (germs incubate up to 2 weeks)
 - Mosquitoes breed in stagnant water
 - No ward-level warning goes out

Result: people fall sick before anyone warns them

DAY 7-12 CASE SURGE

HOSPITALS FILL UP
Fever cases jump; late treatment makes leptospirosis deadly.

BLOOD RUSH
Dengue patients need platelets, which last only 5 days.

WEEKS LATER LATE RESPONSE

CITYWIDE ADVICE
One message for all wards, days later.

OFFICIAL REPORTS
IDSP outbreak data goes online ~6 weeks later.

NEWS-BASED AI
Flags outbreaks only after media reports.

EXISTING SOLUTIONS & THEIR GAPS

WHAT EXISTS

IHIP (govt health portal)

IDSP weekly reports

Health Sentinel (AI news scan)

ARTPARK dengue forecast

BMC advisories

WHAT IT MISSES

Officials only; counts cases already seen

Go online ~6 weeks late

Alerts after people fall sick

Dengue only; Karnataka, not Mumbai

Citywide, days after the flood

Proposed Solution & Key Features

VARSHA: which wards, when, and what to do, before the fever wave

Real-world issue

Floods spread leptospirosis, dengue and malaria. People get just **72 hours** to seek preventive care, but warnings come days later and official data ~6 weeks later.

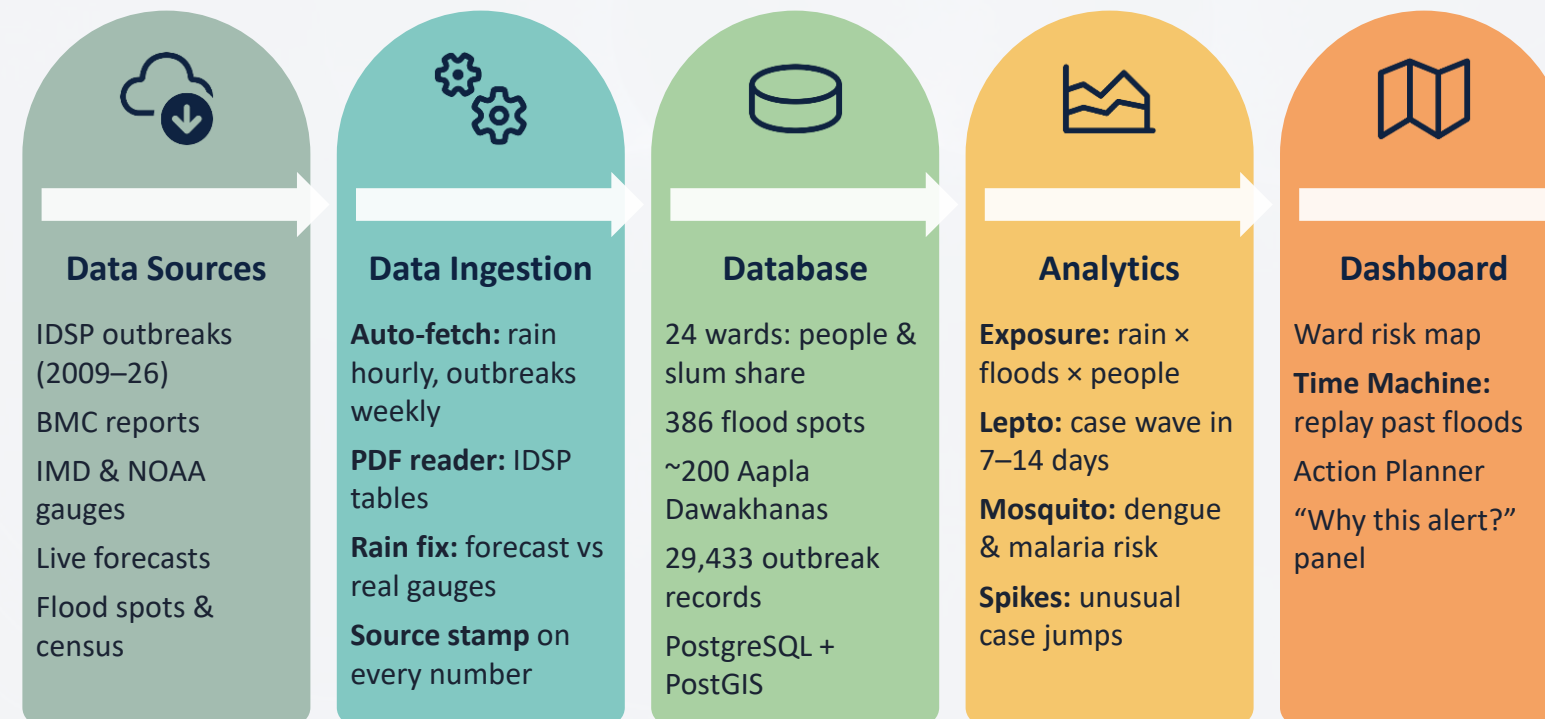
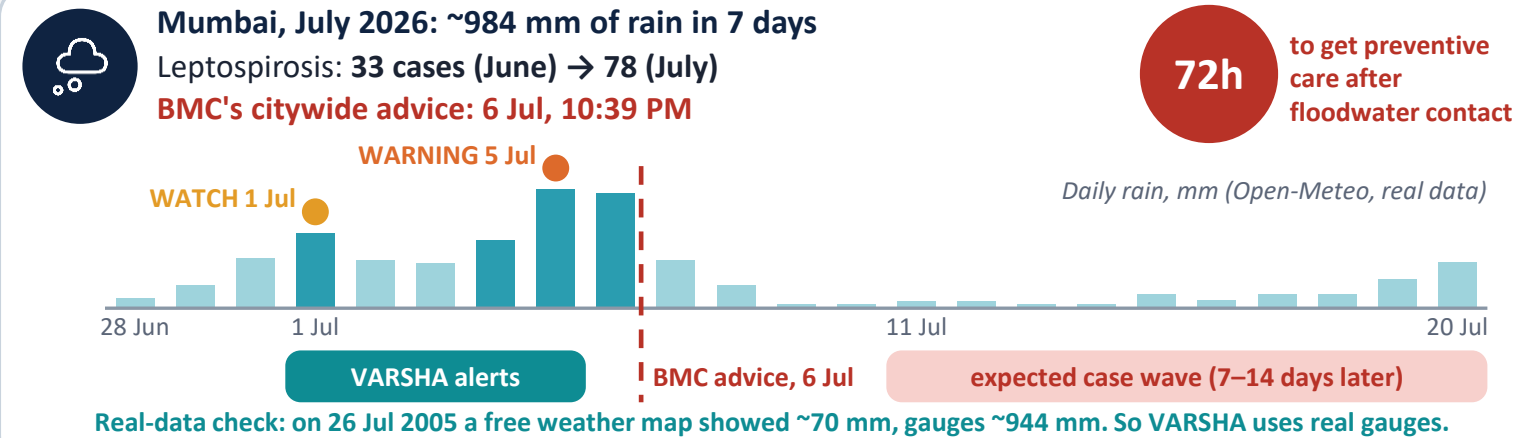
Proposed Solution

VARSHA turns live rainfall into **ward-level disease alerts** for Mumbai. Using **100% real public data** (17 years of IDSP outbreak records, IMD & NOAA rain gauges, live forecasts, BMC flood spots and Census ward data) it predicts where leptospirosis, dengue and malaria will rise **before cases peak**, shows BMC where to act first, and alerts citizens in Marathi, Hindi & English.

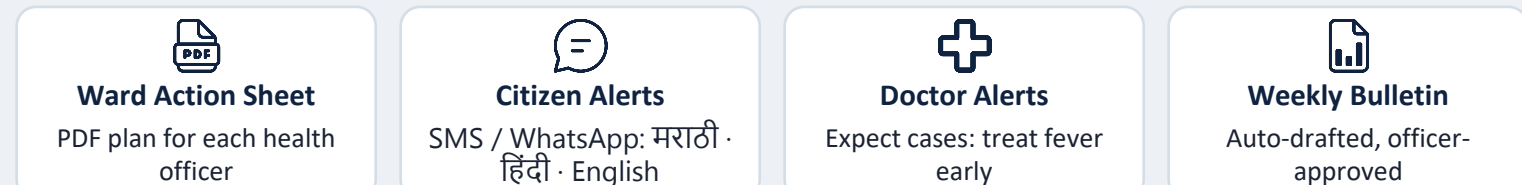
Alert Levels

- NORMAL** No unusual rain · routine checks
- WATCH** Heavy rain (≥ 64.5 mm) on flood-prone wards → clinics on standby
- WARNING** Very heavy rain (≥ 115.6 mm) + flooding → fever camps, SMS alerts
- EMERGENCY** Extreme rain (≥ 204.5 mm) or days of flooding → all-out response

Rain levels = IMD's official 24-hour rainfall categories



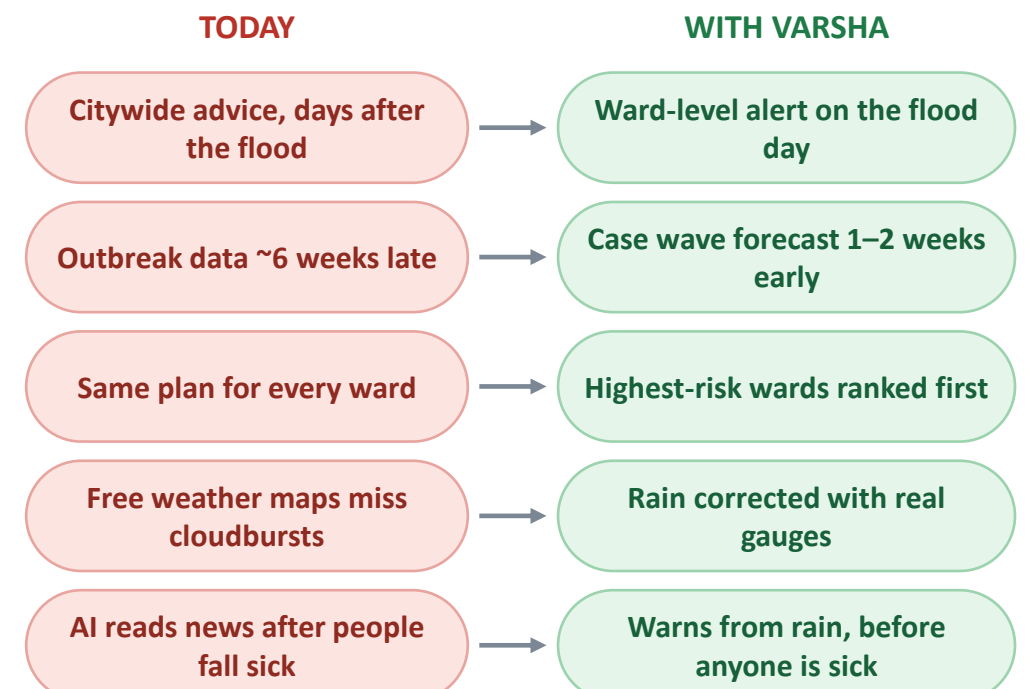
Reports & Alerts



HOW IT ADDRESSES THE PROBLEM

- Same-Day Ward Alerts**
Warns flood-hit wards on the flood day, inside the 72-hour window, not days later.
- Forecast 1–2 Weeks Early**
Predicts when leptospirosis cases will peak, and rates dengue & malaria risk weeks ahead.
- Smart Action Planner**
Sends limited fever camps, doctors and spray teams to the highest-risk wards first.

INNOVATION & UNIQUENESS



Technical Approach & Innovation

Real data in, decisions out. Every number has a source.

Technologies Used



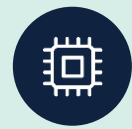
UI & Presentation

React 18 + Vite · Tailwind CSS · Leaflet ward maps · Recharts · English / हिंदी / मराठी toggle



Backend & API

Python 3.11 · FastAPI + Uvicorn · Pydantic v2 · scheduled jobs: rain every hour, outbreak data every week



Intelligence Engine

Pandas, NumPy, SciPy · Statsmodels lag model · LightGBM · R surveillance (EARS, Farrington) · OR-Tools



Data, Hosting & Security

PostgreSQL + PostGIS · Open-Meteo, IMD & NOAA rain data · SHA-256 source stamps · Docker on one server



Links

- Working Prototype** ▶ [\[add your prototype link\]](#)
- Code** ▶ [\[add your GitHub link\]](#)
- Data sources & test log** ▶ [\[add your link\]](#)

Python

FastAPI

React

Leaflet

PostgreSQL + PostGIS

Pandas

Statsmodels

LightGBM

R surveillance

OR-Tools

Docker

Telegram API

Methodology & Process

Mumbai Health-Weather Data Hub

17 years of IDSP outbreak records (29,433 rows) · BMC monsoon disease reports · 24 wards, 386 flood spots, ~200 clinics
Live rain forecasts (Open-Meteo) + real rain gauges (IMD, NOAA) + news (GDELT) · PostgreSQL + PostGIS

Jal-Sanket: Ward Exposure Score

Rain × flood spots × people, for every ward, every day
Rain corrected with real gauges, not raw free maps

Rog-Ghadi: Leptospirosis Forecast

Learns the 7–12 day gap between flooding and new cases
Predicts when the case wave comes and how big, with a range

Machhar-Mapak: Dengue & Malaria Forecast

Uses heat, humidity and rain from the past 2–8 weeks
Rates each ward's mosquito-disease risk weeks ahead

Early-Warning Alert Queue

Combines all models into WATCH / WARNING / EMERGENCY
Ranks all 24 wards by urgency, every hour

Satark: Unusual Spike Detector

Same checks public-health agencies use (EARS, Farrington)
Flags unusual jumps in IDSP & BMC data; news as cross-check

Niyojan: Action Planner

Places camps, doctors & spray teams where they help most
Follows BMC's own monsoon action plan (OR-Tools)

Nagrik Sandesh: Citizen Alerts

Ward-level SMS / WhatsApp in Marathi, Hindi & English
“Waded through floodwater? Visit an Aapla Dawakhana.”

Ward Officer Cockpit

One-page action sheet for every ward health officer
Officer approves alerts; every number shows its source

Impact & Future Scope *Hours, not weeks. Mumbai first, then every district.*

IMPACT

Hours, Not Weeks

Ward alerts go out on the flood day, not days later, and not 6 weeks later like official data.



Inside the 72-Hour Window

People exposed to floodwater are told to get free preventive care while it still works.



Right Ward, First

Fever camps, doctors and spray teams reach the highest-risk wards first, not spread thin.



Protects the Most Vulnerable

Focuses on flood-prone slums, home to about 4 in 10 of Mumbai's 12.4 million people.



Ward Pilot → Full City

Pilot in one flood-prone ward for one monsoon, then scale to all 24 wards with BMC's data.



FUTURE SCOPE

Every District in India

IDSP covers every district, so the same system can protect Chennai, Kolkata, Kerala and Assam.



More Health Risks

The same pipeline can warn about heatstroke, polluted-air illness and water-borne disease.



Blood Banks Ready

Warn blood banks before dengue peaks, since platelets last only 5 days.



POTENTIAL CHALLENGES & SOLUTIONS



CHALLENGE

BMC shares only citywide case totals, not ward-wise numbers.



SOLUTION

Predict ward exposure now; add BMC's ward data later through an official tie-up.



CHALLENGE

Free weather maps miss Mumbai cloudbursts (~70 mm vs ~944 mm, 26 Jul 2005).



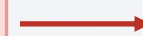
SOLUTION

Correct every forecast with real rain-gauge data (IMD, NOAA) before use.



CHALLENGE

Too many alerts make people ignore them.



SOLUTION

Three clear alert levels, flood-hit wards only, and officer approval first.



CHALLENGE

Health advice must be safe and correct.



SOLUTION

VARSHA never prescribes; it only points people to BMC clinics, as BMC's own advice does.



Supporting Information

A real flood replay, verified data, and the research behind every claim

Case Study: July 2026 Floods

- **1–7 Jul:** ~984 mm of rain at Santacruz
- **4–6 Jul:** red alert; flooding in Andheri, Kurla, Chembur, Powai
- **6 Jul, 10:39 PM:** BMC's citywide advice to get preventive care within 72 hours
- **July:** leptospirosis cases more than doubled (33 → 78)

What VARSHA Would Have Done

Replayed on the real July 2026 rain data, VARSHA's rules give a **WATCH on 1 Jul** and a **WARNING by 5 Jul**, before BMC's citywide advice on the night of 6 Jul. It would then:

- Alert **only flood-hit wards**, in Marathi, Hindi & English
- Send people to an **Aapla Dawakhana within 72 hours**
- Place **fever camps** where risk is highest
- Forecast the **case wave for ~11–20 Jul**

KEY RESEARCH FINDINGS

- a Cases follow floods by 7–12 days**
After the 2005 deluge, leptospirosis patients rose 8×, mostly on days 7–12 (Supe et al., 2018).
- b Official data is ~6 weeks late**
IDSP outbreak reports go online 37–51 days after each week ends (our check of 7 reports, 2025–26).
- c Free weather maps miss cloudbursts**
26 Jul 2005: a free global map shows ~70 mm; real gauges recorded ~944 mm. VARSHA corrects with gauges.

STEP 1 WHAT GOES IN

LIVE RAIN
Hourly forecast for all 24 wards + real gauges (IMD, NOAA).

FLOOD & PEOPLE MAP
386 flood spots, ward slum population, ~200 clinics.

STEP 2 MEASURING

RAIN CORRECTION
Fixes forecast errors using real gauge records.

EXPOSURE SCORE MAP
Rain × flood spots × people, for each ward, each day.

STEP 3 FORECAST

- OUTBREAK FORECAST**
- Leptospirosis: case wave 7–14 days after floods
 - Dengue & malaria: risk weeks ahead
 - Spike check on IDSP & BMC data

Result: WATCH / WARNING / EMERGENCY for every ward

STEP 4 DECISION

ACTION PLANNER
Camps, doctors and spray teams go to the top-risk wards first.

WHY THIS ALERT?
Shows the rain, flood spots and past floods behind it.

STEP 5 ALERTS

CITIZEN ALERTS
“Visit an Aapla Dawakhana within 72 hours.”

WARD ACTION SHEET
One-page plan for each ward health officer.

DOCTOR ALERT
Expect leptospirosis cases; treat fever early.

KEY REFERENCES

OUR CLAIM

Cases rise 7–12 days after floods

Lepto cases 33 → 78 in July 2026

Care within 24–72 hours

29,433 IDSP outbreak records

Dengue warning is “rudimentary”

SOURCE

→ Supe et al., Natl Med J India, 2018

→ BMC data via Free Press Journal, 2026

→ BMC advisory, 6 Jul 2026

→ IDSP (NCDC), via Dataful

→ IITM Pune, Scientific Reports, 2025